Topography Preview

Section 1

Wealth Topography Section

1.1 Introduction to the Wealth Topography Section

The Wealth Topography section includes cross-country data that capture, at the aggregate level, the evolution of household portfolios of assets and debt. The data come from national accounts, household surveys, and a range of research projects and reports.

These data offer a unique view of aggregate household balance sheets. Country-specific portfolios are shaped by, and reflect, national characteristics, including demographic trends, inflation and interest rate dynamics, features of financial and credit markets, the relative importance of stock exchanges versus banking systems, the strength of asset management industries, and the preferential tax treatment of assets, as well as the preferences of households and the generosity of pensions systems.

This chapter introduces the categories of the Wealth Topography database and the construction of each concept. We refer the interested reader to the Technical Documentation of the Wealth Topography for all information related to the treatment of each source of data as well as the country-specific data coverage.

1.2 The `varcode` in the Wealth Topography Section

As described in the general warehouse structure section, for each country-source pair, the `varcode` is a string that identifies a unique time-series by means of five elements. The first element of the `varcode` is a 1-digit code that identifies the Section of the GC Wealth Project, which for the case of Wealth Topography takes the value `p`. Hence, `varcode` takes the following form:

$$\underbrace{p}_{\text{Section}} - \underbrace{BB}_{\text{Sector}} - \underbrace{CCC}_{\text{Variable Type}} - \underbrace{DDDDDD}_{\text{Concept}} - \underbrace{ZZ}_{\text{Section Specific}}$$

1.2.1 `Sector`

The second element of the `varcode` identifies the sector to which the time-series refers. The Wealth Topography database reports aggregate wealth and the composition of assets and liabilities of three institutional sectors: Households, NPISH (non-profit institutions serving households), and Households & NPISH sectors.[1] The classification of institutional sectors used in the Wealth Topography database (and in the GC Wealth Project more in in general) mirrors the classification used by the System of National Accounts 2008 (SNA2008, hereafter) and its European counterpart, the European System of Accounts 2010 (ESA2010. hereafter). The institutional sectors are reported in Table 1.1, where the column "Code" reports the sector identifier used in the `varcode` while the column "Description" reports a detailed, non-technical, description of each sector.

[1] Other institutional sectors (non-financial corporations sector, financial corporations sector, general government sector, and the rest of the world) are not currently included in the database.

Table 1.1: Wealth Topography: Sector

Code	Sector	Description
hs	Households	The household sector gathers together all individuals or group of individuals who live together, who pool their income and wealth, and make joint economic decisions (e.g., consumption). The household sector is one of the institutional sectors constituting the national economy.
hn	Households & NPISH	The households and NPISH sector (non-profit institutions serving households) gathers together households (all individuals or group of individuals who live together, who pool their income and wealth, and make joint economic decisions) and non-profit organizations that primarily provide services to households (such as charities, religious institutions, and social clubs). The household and NPISH sector is one of the institutional sectors constituting the national economy.
np	NPISH	The NPISH (non-profit institutions serving households) sector gathers together non-profit organizations that primarily provide services to households. This includes organizations such as charities, religious institutions, and social clubs. The NPISH sector is one of the institutional sectors constituting the national economy.

1.2.2 `Variable Type`: Consolidation

The third element of the `varcode` identifies the `Variable Type` which, in the Wealth Topography database, coincides with the consolidation status. The Wealth Topography reports outstanding levels of assets and liabilities and net wealth for the institutional sectors considered. In the SNA2008, outstanding levels or stocks for a given sector are obtained by aggregating the accounts of the units of the sector, and the aggregation process can differ according to the treatment of transactions within the sector. Within-sector transactions can be consolidated or non-consolidated accounts. Consolidation refers to the practice of eliminating any transaction between units or sub-sectors within the same sector. In this case, any remaining transaction is a transaction with other institutional sectors of the economy. For example, in a consolidated financial account, a loan between two households within the household sector would be eliminated, and the remaining loans would reflect transactions with other sectors of the economy (e.g., the rest of the world, the government sector, the non-financial corporations sector, or the financial corporations sector). In contrast, in a non-consolidated account, such an elimination is not carried out and the aggregation across units within the same sector includes all types of transactions, regardless of the counterpart. In this case, a loan between two households within the households sector would be counted as both an asset and a liability. In practice, most statistical agencies publish only non-consolidated accounts because the construction of consolidated accounts requires knowing the sender and the recipient of all transactions.

Table 1.2 reports under the column "Code" the consolidation status identifier used in the `varcode` of the Wealth Topography database while the column "Description" reports a detailed, non-technical description. If a source in the Wealth Topography database does not provide any information about the consolidation status, we classify them simply as "Aggregate".

Table 1.2: Wealth Topography: Variable Type

Code	Consolidation status	Description
agc	Aggregate (Consolidated)	Aggregate value for a given institutional sector for which transactions that occur between units within the same sector are eliminated.
agn	Aggregate (Non-consolidated)	Aggregate value for a given institutional sector for which transactions that occur between units within the same sector are not eliminated. This may result in double-counting of assets and liabilities
agg	Aggregate	Aggregate value for a given institutional sector. The value is reported without knowing whether transactions that occur between units within the same sector are eliminated or kept during the aggregation.

1.2.3 Concept

The fourth element of the `varcode` identifies the `Concept`. In the Wealth Topography database, a concept is a macro-category of the balance sheet, computed by aggregating different asset and liability classes. The construction of such concepts and the aggregation procedure are explained in Section 1.3. The concepts and the corresponding identifiers used in the `varcode` of the Wealth Topography database, together with a detailed description, are reported in Table 1.3.

Table 1.3: Wealth Topography: Concept

Code	Concept	Description
netwea	Net Wealth	Net wealth is the sum of all tangible assets (real estate properties and land, valuables, plants, machineries, equipment, etc.) and intangible assets (stocks, bonds, balances of current, saving, and investment accounts, private pension and life insurance funds, etc.) minus the sum of all debts and liabilities (mortgages, loans, and credit card debt).
nnhass	Financial Assets & Fixed Capital of Personal Businesses	Sum of all financial assets (such as stocks, shares or equities in corporations and quasi-corporations, corporate and government bonds, mutual funds, cash, current, savings and investment accounts, time deposits or certificates of deposit (CDs), accumulated balance in private pension funds, cash or reserve value of life insurance funds, etc.) and the fixed capital stock of small personal businesses of producer households (such as plants, machinery, equipment, inventories, software, and goodwill).
fliabi	Debt	Debts, or liabilities, are financial obligations that generally result from borrowing. For households, the most common form of debt is the mortgage, a loan taken out by an individual or a household to purchase a home or other real estate properties. Other forms of debt include credit card debts, auto loans, and student loans.
fliabm	Debt secured by a mortgage	Debt secured by a mortgage are loans taken out by an individual or a household to purchase a home or other real estate properties.
facdbl	Cash, Deposits, Bonds & Loans	Cash can be held in the form of banknotes, coins, or digital currency. Bonds represent a loan made by households to a corporation or government entity and provide a fixed return to the investor in the form of interest payments. Deposits include both current accounts (available on demand) and savings accounts (generally interest-bearing instruments with limited withdrawal). Loans are the money lent to individuals or households for which they owe the principal and interest to the creditors.
fadepo	Deposits	Deposits include both current accounts (available on demand) and savings accounts (generally interest-bearing instruments with limited withdrawal).
faeqfd	Stocks, Business Equities & Fund Shares	Stocks (or shares) and business equities are securities representing the ownership in a corporation or quasi-corporation and entitling the holder to a portion of the company's profits. Fund shares represent the participation in mutual or investment funds which are investment vehicles that pool together money from multiple investors to purchase a portfolio of securities such as stocks, bonds, or other assets.
falipe	Pensions & Life Insurance	Private pension assets refer to funds that are set aside by an individual (or an employer) to provide income during retirement (excluding public pension and social security schemes). Pension assets can take the form of defined benefit plans (retirement income benefits are fixed) or defined contribution plans (retirement income benefits are dependent on contributions). Life insurance assets refer to the cash value or reserve value of a life insurance policy (where the policyholder pays premiums in exchange for a death benefit that is paid out to their beneficiaries upon their death).
nfabus	Fixed Capital of Personal Businesses	Fixed capital of small personal businesses represents the non-financial assets that are used by a small personal business to produce goods or services and to support the ongoing operations of the business. This category includes both tangible assets (such as plants, machineries, equipment, inventories, valuables) and intangible assets (such as software and goodwill).
nfahou	Housing & Land	Housing identifies the value of residential buildings or structures that are owned by households (may include single-family homes, apartments, condominiums, and other types of housing). Land assets include agricultural land, residential land, commercial land, and industrial land. This includes land underlying dwellings, and other buildings and structures.

nfhous	Housing	Housing identifies the value of residential buildings or structures that are owned by households (may include single-family homes, apartments, condominiums, and other types of housing).
nfalan	Land	Land assets include agricultural land, residential land, commercial land, and industrial land. This includes land underlying dwellings, and other buildings and structures.
offsho	Total Financial Offhore Wealth	Offshore financial wealth refers to the equities, bonds, mutual fund shares, and associated bank deposits owned by households in banks outside of their country of residency.
nfadur	Consumer Durable Goods	A durable good is a consumption good that can deliver useful services to a consumer through repeated use over an extended period of time

We use a stylized balance sheet to show the relationship between the concepts in Table 1.3. The balance sheet is organized into two panels: Assets, and Liabilities and Net Wealth. We also report illustrative values for each concept. Among assets, “Housing & Land” and “Financial Assets & Fixed Capital of Personal Businesses” are the two main concepts. “Housing & Land” is the sum of the concepts “Housing” and “Land”. The concept “Financial Assets & Fixed Capital of Personal Businesses”, in turn, equals the sum of the following concepts: “Fixed Capital of Personal Businesses”, “Cash, Deposits, Bonds & Loans”, “Stocks, Business Equities & Fund Shares”, and “Pensions & Life Insurance”. “Deposits” is a part of “Cash, Deposits, Bonds & Loans”, but also a concept on its own. We also include additional concepts “Offshore Financial Wealth” and “Consumer Durable Goods”. The concept “Debt” constitutes the liability side of the balance sheet. As mortgage debt is the key component of debt in many countries, the concept “Debt Secured by a Mortgage” provides data on debt at an additional level of disaggregation. Finally, the concept “Net Wealth” is the balancing item and it equals the difference between total assets (“Housing & Land” and “Financial Assets & Fixed Capital of Personal Businesses”) and “Debt”. Note that we follow the standard definition of net wealth and, therefore, do not include offshore financial wealth or consumer durable goods in the net wealth definition. That is why we separate the two concepts in the right panel of the balance sheet.

Table 1.4: Wealth Topography: Balance Sheet Example

Assets		Liabilities and Net Worth	
Housing & Land	1510		
Housing	800		
Land	710		
Financial Assets & Fixed Capital of Personal Business	3518		
Fixed Capital of Personal Businesses	26		
Cash, Deposits, Bonds & Loans	684		
Deposits	350		
Stocks, Business Equities & Fund Shares	1550		
Pensions & Life Insurance	1258		
Offshore Financial Wealth	241		
Consumer Durables	278		
		Debt	695
		Debt Secured by a Mortgage	400
		Net Wealth	4333
		Net Wealth incl. Offshore Wealth and Consumer Durable Goods	4852

1.2.4 `Section-Specific`: Recording of Financial Position

The fifth element of the `varcode` is a `Section-Specific` string which, in the Wealth Topography databse, corresponds to the financial position of each concepts. Financial assets may be recorded gross or net of all liabilities, at a given point in time. Similarly, debt securities can appear both as assets or as liabilities. To distinguish between the financial position of each concept, the Wealth Topography distinguishes between three types of financial position: gross assets (`ga`), net assets (`na`), and liabilities (`lb`). When this classification is not possible, the financial position is not applicable (`_`). Table 1.5 provides a taxonomy of financial positions together with a non-technical description for each.

Table 1.5: Wealth Topography: Section-Specific

Code	Financial position	Description
ga	Gross Assets	Gross assets are the total value of all assets before subtracting the value of debts and liabilities.
na	Net Assets	Net assets are the total value of all assets after subtracting the value of debts and liabilities.
lb	Liabilities	Liabilities are financial obligations or debts that an individual, company, or organization owes to another party.
___	Not Applicable	Not Applicable

1.3 Construction of the Wealth Topography Database

We now give a brief and stylized description of the process underlying the construction of the Wealth Topography database. In its most stylized version, this process involves three steps (synthesized in the flow chart):

1. Identify and download raw data for each source.
2. Map every raw source into national accounting concepts using the Wealth Topography Conceptual Grid.
3. Create Wealth Topography macro-category ('concept') by aggregating the variables from each source (transformed following step 2) using the Wealth Topography Composition Rules.

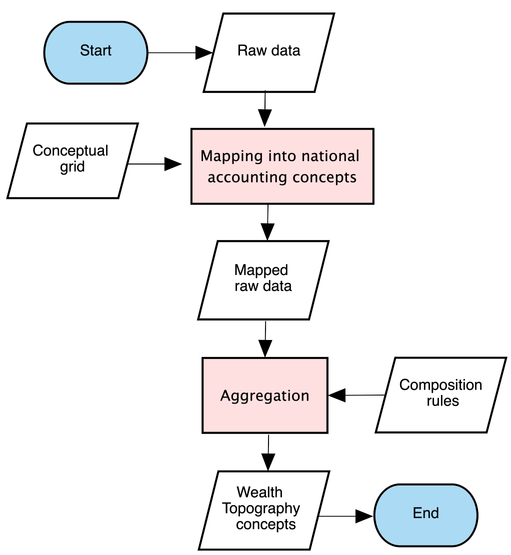

Figure 1.1: Wealth Topography: Flow Chart

We now provide a detailed explanation of all steps put in place to construct the Wealth Topography database.

1.3.1 Identify and Download Raw Data for Each Source

The starting point is the identification of raw data on balance sheets from various sources, such as national accounts, household surveys, and a range of research projects and reports. Once the sources have been identified and the raw data downloaded, we classify them in three macro groups:

1. Raw data sources published by central banks and national statistical institutes that use the SNA2008/ESA2010 framework to organize and disseminate the data. We refer to these raw data sources simply as cross-national official statistics.
2. Raw data sources published by central banks and national statistical institutes that use variants of the SNA2008/ESA2010 framework or other frameworks to organize and disseminate the data. We refer to these raw data sources as cross-national official statistics that use variants of SNA2008/ESA2010 framework.
3. Raw data sources contained in surveys and academic papers or published by central banks and research institutes that use frameworks different from the SNA2008/ESA2010 framework to organize and disseminate the data. We refer to these data sources as cross-national official survey data or cross-national academic research that do not use the SNA2008/ESA2010 framework.

1.3.2 Map Raw Data into National Accounting Concepts

Once raw data have been obtained and classified, we harmonize them using the Wealth Topography Conceptual Grid. This grid, inspired by National Accounts, assigns an alphanumeric identifier to each item or instrument of the balance sheet.[2] An extract of the grid can be seen in Table 1.6. Under the column "Code", we report the code that identifies each item of the balance sheet (e.g., AN111 stands for Dwellings). The code consists of two elements (letters and numbers) and a special character for the case of financial assets and liabilities. The alphabetic part of the code identifies whether a specific balance sheet component falls in the category of non-financial assets (AN), financial assets (A_AF), or liabilities (L_AF), while numbers identify the class of assets and liabilities to which the instrument belongs.

Equipped with the grid, we can map each raw source into national accounting concepts with the aim of obtaining harmonized data on net wealth, assets, and liabilities that can be compared across countries and sources. According to the type of raw data source, we distinguish between two types of mapping:

- Automatic mapping (for cross-national official statistics). For raw data sources published by central banks and national statistical institutes that use the SNA2008/ESA2010 framework to organize and disseminate the data, the mapping of raw data into national accounting concepts is straightforward.
- Conceptual or manual mapping (for cross-national official statistics that use variants of SNA2008/ESA2010 framework). For raw data sources published by central banks and national statistical institutes that use variants of the SNA2008/ESA2010 framework or other frameworks to organize and disseminate the data, the mapping of raw data into national accounting concepts is not straightforward. Therefore, we manually map each series from the raw data source to elements of the grid. The mapping is conceptual and based on our reading of source-specific documentation. All source-specific manual mapping tables are reported in Appendix **??**.

For cross-national official survey data or cross-national academic research that do not use the SNA2008/ESA2010 framework, we build source-specific grids to help cross-walking from raw data to the national accounting concepts in Table 1.6. For completeness, we report the source-specific grids together with source-specific documentation in Appendix **??**. We followed the SNA2008/ESA2010 definition for consumer durable goods and debt secured by a mortgage. For offshore financial wealth, we created a framework which is analogous to the financial wealth definition.

[2]For the original table used in ESA2010 @ESA2010, see Section 'IV.3: Balance sheets: Closing balance sheet' in 'Table 24.6 — Full sequence of accounts for households' of the European System of National Accounts accessed via https://ec.europa.eu/eurostat/esa2010/.

Table 1.6: Wealth Topography: Conceptual Grid

Code	Description	Financial position
AN	Produced and non-produced non-financial assets	ga
AN1	Produced non-financial assets	ga
AN11	Fixed assets by type of assets	ga
AN111	Dwellings	ga
...	...	...
...	...	...
A_AF	Financial assets	ga
A_AF1	Monetary gold and SDRs	ga
A_AF11	Monetary gold	ga
A_AF12	SDRs	ga
A_AF2	Currency and deposits	ga
A_AF21	Currency	ga
...	...	...
L_AF	Liabilities	lb
...	...	...
L_AF3	Debt securities	lb
L_AF31	Short-term debt securities	lb
L_AF32	Long-term debt securities	lb
L_AF4	Loans	lb
L_AF41	Short-term loans	lb
L_AF42	Long-term loans	lb
...	...	...
A_AXF	Financial assets held offshore	ga
XDHHCE	Consumer durables	ga

1.3.3 Create Wealth Topography Concepts Using Our Composition Rules

For raw sources classified as cross-national official statistics (independently of whether the SNA2008/ESA2010 framework is used), the mapping yields harmonized series on net wealth, assets, and liabilities. The next step is to aggregate these harmonized raw data into the macro-categories (`concept`) of the Wealth Topography database. Aggregation is carried out through the general composition rules in Table 1.7.

In practice, we use an extended set of composition rules relative to those displayed in Table 1.7, since different data sources provide different levels of aggregation for the sub-components of the balance sheet. For example, the `concept` "Stocks, Business Equity & Fund Shares" can be computed by aggregating different variables. On the one hand, for some data sources, the `concept` "Stocks, Business Equity & Fund Shares" is set equal to "Equity and investment fund shares (A_AF5)". On the other hand, when the aggregate variable is missing, we obtain "Stocks, Business Equity & Fund Shares" by summing up "Equity (A_AF51)" and "Investment fund shares/units (A_AF52)". We always provide a detailed account of which composition rule has been used to construct a data point in the Wealth Topography database in the metadata section of the warehouse file or graphs available to users through the website of the GC Wealth Project. All source-specific composition rules used in the Wealth Topography database are reported in the source-specific documentation.

Finally, for raw data sources classified as cross-national official survey data or cross-national academic research that do not use the SNA2008/ESA2010 framework, we consider ad-hoc source-specific composition rules. In this case, raw data sources are not harmonized by mapping them into national accounting concepts using the grid. Rather, based on our reading of source-specific documentations, the aggregation of this raw data type uses macro-categories that are coherent and comparable to those obtained from cross-national official statistics. We provide a detailed treatment of these sources in Appendix ??.

1.4 Additional Information on the Wealth Topography Section

For any additional information about the Wealth Topography database the reader can refer to Appendix **??** which includes a list of all sources covered by the Wealth Topography database, an extensive treatment of the composition rules, and detailed source-specific documentations.

Table 1.7: Wealth Topography: General Composition Rule

Code	Concept	Composition rule using codes	Composition rule
netwea	Net Wealth	(AN1) + (AN2) + (A_AF) - (L_AF)	(Produced non-financial assets) + (Non-produced non-financial assets) + (Financial assets) - (Liabilities)
nnhass	Financial Assets & Fixed Capital of Personal Businesses	(AN1) + (AN2) + (A_AF) - (AN111) - (AN112) - (AN211)	(Produced non-financial assets) + (Non-produced non-financial assets) + (Financial assets) - (Dwellings) - (Other buildings and structures) - (Land)
fliabi	Debt	(L_AF3) + (L_AF4) + (L_AF5) + (L_AF6)	(Debt securities, liab.) + (Loans, liab.) + (Equity and investment fund shares, liab.) + (Insurance, pension and standardized guarantee schemes, liab.)
fliabm	Debt secured by a mortgage	(XAF42LM)	(Long-term loans secured by a mortgage, liab.)
facdbl	Cash, Deposits, Bonds & Loans	(A_AF2) + (A_AF3) + (A_AF4)	(Currency and deposits) + (Debt securities) + (Loans)
fadepo	Deposits	(A_AF22) + (A_AF29)	(Transferrable deposits) + (Other deposits)
faeqfd	Stocks, Business Equities & Fund Shares	(A_AF5)	(Equity and investment fund shares)
falipe	Pensions & Life Insurance	(A_AF6)	(Insurance, pension and standardized guarantee schemes)
nfabus	Fixed Capital of Personal Businesses	(AN1) + (AN2) - (AN111) - (AN112) - (AN211)	(Produced non-financial assets) + (Non-produced non-financial assets) - (Dwellings) - (Other buildings and structures) - (Land)
nfahou	Housing & Land	(AN111) + (AN112) - (AN1123) + (AN21111)	(Dwellings) + (Other buildings and structures) + (Land)
nfalan	Land	(AN211)	(Land underlying dwellings) + (Land underlying other buildings and structures) + (Land under cultivation)
nfhous	Housing	(AN111) + (AN112)	(Dwellings) + (Other buildings and structures)
offsho	Offshore Financial Wealth	(A_AXF)	(Financial assets held offshore)
nfadur	Consumer Durable Goods	(XDHHCE)	(Consumer durables)